23Fall MAT1011 Midterm with Answers

MathPi

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Question 1

(a) For $x \neq 0$,

$$f'(x) = \frac{1}{3}(8x-5)x^{2/3}.$$

At $x = 0$,

$$f'(0) = \lim_{h\to 0} \frac{(h-1)h^{5/3}}{h} = \lim_{h\to 0}(h-1)h^{2/3} = 0.$$

Thus the formula for f' holds for every $x \in \mathbb{R}$. For $x \neq 0$,

$$f''(x) = \frac{10}{9}(4x-1)x^{-1/3}.$$

Since

$$\frac{f'(h)-f'(0)}{h} = \frac{8h-5}{3h^{1/3}}$$

has no finite limit as $h \to 0$, $f''(0)$ does not exist.

(b) Since $f(0) = f'(0) = 0$, the tangent line and normal line are

$$y = 0 \quad \text{and} \quad x = 0, \text{ respectively.}$$

(c) The derivative exists everywhere, and

$$f'(x) = 0 \iff (8x-5)x^{2/3} = 0.$$

Hence the critical numbers are

$$x = 0, \qquad x = \frac{5}{8}.$$

The corresponding points are

$$(0,0), \qquad \left(\frac{5}{8}, -\frac{3}{8}\left(\frac{5}{8}\right)^{5/3}\right).$$

Question 2

(a) Using $\sin t \sim t$ as $t \to 0$, we have

$$\sin^3(\sin^2 x) \sim x^6, \qquad \sin^2(\sin(x^3)) \sim x^6.$$

Therefore,

$$\lim_{x\to 0} \frac{\sin^3(\sin^2 x)}{\sin^2(\sin(x^3))} = 1.$$

(b) We write

$$\frac{\cos(1+\sqrt{x})-\cos 1}{\sqrt{\sin x}} = \frac{\cos(1+\sqrt{x})-\cos 1}{\sqrt{x}}\sqrt{\frac{x}{\sin x}}.$$

The two factors tend to $-\sin 1$ and 1, respectively. Hence

$$\lim_{x\to 0^+} \frac{\cos(1+\sqrt{x})-\cos 1}{\sqrt{\sin x}} = -\sin 1.$$

(c) Let

$$F(u) = \int_0^u \left[\cos(t^2) + \cos^2(t^2)\right] dt.$$

By the Fundamental Theorem of Calculus, $F(0) = 0$ and $F'(0) = 2$. Thus

$$\lim_{x\to 0} \frac{F(x^2)}{x^2} = F'(0) = 2.$$

Question 3

Rationalizing gives

$$\lim_{x\to 0} \frac{\sqrt{1+2x}-1}{x} = \lim_{x\to 0} \frac{2}{\sqrt{1+2x}+1} = 1.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{2}\right\}.$$

If $0 < |x| < \delta$, then $1 + 2x > 0$ and

$$\begin{aligned}\left|\frac{\sqrt{1+2x}-1}{x} - 1\right| &= \left|\frac{2}{\sqrt{1+2x}+1} - 1\right| \\ &= \frac{2|x|}{\left(\sqrt{1+2x}+1\right)^2} \\ &\le 2|x| < \varepsilon.\end{aligned}$$

This proves the limit by the ε-δ definition.

Question 4

(a) Integration by parts on each side of 0 gives

$$\int |x|\cos x\, dx = \begin{cases} x\sin x + \cos x + C, & x \ge 0, \\ -x\sin x - \cos x + C + 2, & x < 0. \end{cases}$$

The constants are chosen so that the two branches agree at 0. The resulting function is differentiable at 0, with derivative 0.

(b) Integration by parts yields

$$\begin{aligned} J_n &= \int \sin^{n-1} x \sin x\, dx \\ &= -\sin^{n-1} x\cos x + (n-1)\int \sin^{n-2} x\cos^2 x\, dx \\ &= -\sin^{n-1} x\cos x + (n-1)(J_{n-2} - J_n). \end{aligned}$$

Rearranging, we obtain

$$J_n = \frac{n-1}{n} J_{n-2} - \frac{1}{n} \sin^{n-1} x \cos x, \qquad n \geq 2,$$

with constants of integration chosen consistently.

(c) By the Fundamental Theorem of Calculus,

$$\int \left[\frac{d}{dx} f(\cos^2 x) \right] dx = f(\cos^2 x) + C,$$

whereas

$$\frac{d}{dx} \int f(\cos^2 x)\, dx = f(\cos^2 x).$$

Therefore the required expression is

$$C, \qquad C \in \mathbb{R}.$$

Question 5

(a) The answer is **YES**. Since $u(0) = 0$ and $|D(x)| \leq 1$,

$$\left| \frac{u(x) - u(0)}{x} \right| \leq \left| \frac{g(x) - g(0)}{x} \right| \longrightarrow |g'(0)| = 0.$$

Hence

$$u'(0) = 0.$$

(b) The answer is **YES**. We prove Riemann integrability using upper and lower sums. Since $0 \leq \zeta(x) \leq 1$,

$$0 \leq 1 - f(x) = 1 - \cos(\zeta(x)) \leq \frac{\zeta(x)^2}{2}, \qquad 0 \leq f(x) \leq 1.$$

Given $\varepsilon > 0$, choose a positive integer Q such that

$$\frac{1}{2Q^2} < \frac{\varepsilon}{2}.$$

The set

$$E_Q = \left\{ \frac{p}{q} \in [0,1] : \gcd(p,q) = 1,\ 1 \leq q \leq Q \right\}$$

is finite. Write $M = |E_Q|$. For $x \notin E_Q$,

$$0 \leq 1 - f(x) \leq \frac{1}{2Q^2}.$$

Let P_m be the uniform partition of $[0,1]$ into m subintervals. At most $2M$ of these subintervals meet E_Q. Their total length is at most $2M/m$, and the oscillation of f on each remaining subinterval is at most $1/(2Q^2)$. Consequently, for sufficiently large m,

$$U(P_m, f) - L(P_m, f) \leq \frac{2M}{m} + \frac{1}{2Q^2} < \varepsilon.$$

Thus f is Riemann integrable. Every nondegenerate subinterval contains an irrational number, at which $f(x) = 1$, so $U(P_m, f) = 1$. It follows that

$$\int_0^1 \cos(\zeta(x))\, dx = 1.$$

Question 6

Since f' is continuous and never zero on $[0,1]$, it has a constant sign. Therefore f is strictly monotone.

(a) By the mean value theorem for integrals, there exists $\xi_* \in [0,1]$ such that

$$\int_0^1 f(x)\,dx = f(\xi_*).$$

Strict monotonicity of f implies that ξ_* is unique.

(b) Let

$$m_0 = \min_{x\in[0,1]} f(x), \qquad M_0 = \max_{x\in[0,1]} f(x).$$

Since S_n is the average of n values of f,

$$m_0 \le S_n = \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) \le M_0.$$

By continuity, $f([0,1]) = [m_0, M_0]$, so there exists $\xi_n \in [0,1]$ such that

$$S_n = f(\xi_n).$$

Strict monotonicity implies uniqueness.

(c) Since S_n is a Riemann sum for the continuous function f,

$$S_n \longrightarrow I := \int_0^1 f(x)\,dx = f(\xi_*).$$

By continuity and nonvanishing of f',

$$c := \min_{x\in[0,1]} |f'(x)| > 0.$$

The mean value theorem gives

$$c|\xi_n - \xi_*| \le |f(\xi_n) - f(\xi_*)| = |S_n - I|.$$

Given $\varepsilon > 0$, choose a positive integer $N = N(\varepsilon)$ such that $|S_n - I| < c\varepsilon$ whenever $n > N$. Then

$$|\xi_n - \xi_*| \le \frac{|S_n - I|}{c} < \varepsilon \qquad \text{whenever } n > N.$$